

Technical Specification: E-commerce Profitability Analysis Project

1. Objective

Analyze e-commerce sales data to identify key drivers of revenue and profitability across products, channels, and pricing strategies.

2. Data Sources

- 1 Amazon Sales Report
- 2 International Sale Report
- 3 Cloud Warehouse Comparison
- 4 Expense IIGF
- 5 May-2022 & P&L; March 2021
- 6 Sale Report

3. Data Model

- 1 sales (transactions, revenue, qty, channel)
- 2 products (SKU, category, size, stock)
- 3 prices (SKU, platform, price)
- 4 expenses (cost data)
- 5 channels_profit (channel performance)

4. Key Analytical Tasks

- 1 Revenue trends over time
- 2 Profit and margin analysis
- 3 Channel performance comparison
- 4 Pricing differences across platforms
- 5 Top and underperforming products
- 6 B2B vs B2C analysis

5. SQL Requirements

- 1 JOIN operations

- 2 GROUP BY aggregations
- 3 CTEs (WITH)
- 4 Window functions (ROW_NUMBER, SUM OVER)

6. Dashboard Requirements

- 1 KPIs: Revenue, Profit, Orders, AOV
- 2 Time series charts
- 3 Category and SKU analysis
- 4 Channel comparison
- 5 Interactive filters

7. Expected Deliverables

- 1 Cleaned and structured dataset
- 2 SQL scripts
- 3 Dashboard (Power BI/Tableau)
- 4 Insights and business recommendations

8. Success Criteria

- 1 Clear business insights
- 2 Proper data modeling
- 3 Advanced SQL usage
- 4 Professional dashboard